

YASHARTH SINGH

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WORK EXPERIENCE

Trainee | Internship

Jun 2025 - Jul 2025

HCL Infotech Ltd

Noida

- **Engineered** an enterprise intrusion detection system using YOLOv8x and OpenCV for real-time security monitoring, Implemented WebSocket-based alerting with Twilio and SMTP, along with persistent analytics using SQLite.
- **Developed** multi-tenant utility management platform with **Next.js** and **PostgreSQL**, implementing JWT authentication and processing complex tariff calculations with sub-100ms API response times under concurrent loads.
- **Implemented** computer vision pipeline processing **1000+** detections in **<3 seconds** using YOLOv8x, NumPy, and OpenCV; created Flask APIs with session management and multi-format export capabilities.

EDUCATION

Jaypee Institute of Information Technology

B.Tech CSE

PROJECTS

Signalist - Real-Time Stock Analytics Platform | 🐙 Github

Jan 2026 - Present

- **Built** high-performance stock tracking dashboard monitoring 10,000+ securities with sub-200ms latency using Next.js 15 server components, TradingView widgets, and Finnhub API integration.
- **Established** event-driven architecture with Inngest for 1000+ concurrent automated workflows, including price alerts, email notifications via Gemini AI, and MongoDB-backed data persistence with optimized indexing.
- **Engineered** scalable backend handling real-time WebSocket connections and asynchronous job processing, demonstrating expertise in building performant financial technology applications.

Technologies / Tools Used : Next.js 15, TypeScript, MongoDB, Inngest

Image-Caption-Generator | 🐙 Github

Oct 2025

- **Architected** production-grade CNN-LSTM hybrid architecture using VGG16 feature extraction and LSTM decoder, achieving 47% BLEU-1 score on Flickr8k dataset with 8,092 images and 40,000+ captions.
- **Enhanced** training pipeline with custom TensorFlow data generators, reducing memory consumption by 90% (10GB to <1GB) and cutting training time from 25+ hours to 8-9 hours through efficient batch processing.
- **Deployed** Streamlit web application with real-time inference supporting 64 samples/batch on Kaggle P-100 GPU infrastructure, demonstrating ML model deployment and optimization skills

Technologies / Tools Used : VGG16, LSTM, TensorFlow, Keras, Streamlit

Sahyatri – AI Public Transit Occupancy System | 🐙 Github

Apr 2025

Winner, BITBOX 5.0 (Google Developers Group)

- **Designed** edge AI system on Raspberry Pi 4 with fine-tuned YOLOv5, achieving **99% detection accuracy** for real-time passenger counting across varying conditions.
- **Constructed** FastAPI microservices with streamlined inference pipeline, reducing latency by **50%** and enabling real-time streaming to **React.js** dashboard for transit operators.

Technologies / Tools Used : OpenCV, Next.js, Postgres SQL, Raspberry Pi, Yolo, Flutter

SKILLS

Programming Languages :	C, C++, JavaScript, TypeScript, Python, PHP
Frameworks & Libraries :	React, Flask, Next.js, Express, FastAPI, Node.js, Streamlit
Tools & Platforms :	Docker, Git, GitHub, GitLab, Nginx, Cloudflare, Inngest
Databases :	PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis, PrismaORM

AWARDS & ACHIEVEMENTS

Winner BITBOX 5.0

Apr 2025

Google Developers Group - IIIT 128

LEADERSHIP & ACTIVITIES

- Coordinator, Creativity and Innovation cell in Robotics (CICR), IIIT – contributed to robotics and AI-based projects and technical events.
- Collaborated in cross-functional teams during hackathons and academic projects, contributing to system design and implementation.